Exp-1 Basic networking commands in Linux and Windows.

Aim:

To study and execute basic networking commands available in Linux and Windows operating systems for diagnosing and troubleshooting networking issues.

Procedure:

1. Open command prompt (Windows) or Terminal (Linux).

2. Execute 'ipconfig' (Windows) or "config' /ip a' (Linux) to view IP configuration.

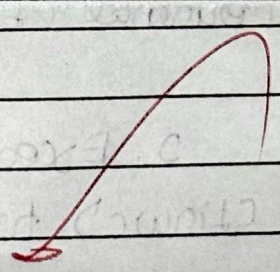
3. Use 'ping 8.8.8.8' to test connectivity to Google's DNS server.

4. Execute 'tracert google.com' (Windows) or 'traceroute google.com' (Linux) to view packet path.

5. Run 'netstat -an' to list active network connections and listening ports.

6. Use 'nslookup google.com' to perform DNS lookup.

```
ipconfig /all          # Windows - Full IP configuration
ip a                   # Linux - IP address info
ping -c 4 8.8.8.8      # Linux - Ping 4 packets
ping 8.8.8.8           # Windows - Ping
traceroute google.com  # Linux - Trace route
```



8. Execute 'arp -a' to view the ARP cache.

9. Record all outputs and analyze the results.

Result:

Basic networking commands were successfully executed on both Linux and commands. IP addresses, routing tables, ARP cache, and DNS resolution were observed and recorded.